M1S1 | Base Recap

Basic Concepts

Indentation

Python uses indentation to define blocks of code

Data Types

int, float, bool, str, list, tuple, set, dict

Operators

+, -, *, /, //, %, **, @, =, +=, ^=, >, or, and, is

Conditionals, Loops, Funcs

- if, elif, else
- for, while
- def

Mutability

each object has type, id, value and methods

```python
x = 5

print(type(x))
print(id(x))
print(x)
print(dir(x))
```

Run main

```
/Users/a89088/PycharmProjects/PythonProject/.venv/b
<class 'int'>
4333707760
5
['__abs__', '__add__', '__and__', '__bool__', '__ce

Process finished with exit code 0
```

Mutable vs Immutable Objects

```python
lst = [1, 2, 3]
print(id(lst))

lst.append(5)
print(id(lst))
```

```
Run    main ×

/Users/a89088/PycharmProjects/PythonProject/.venv/bi
4374088064
4374088064
```

Mutables

with editing current object you do NOT create the new one: list, dict, set

```python
x = 5
print(id(x))

x += 2
print(id(x))
```

```
Run    main ×

/Users/a89088/PycharmProjects/PythonProject/.venv/bin/python /U
4318978544
4318978608

Process finished with exit code 0
```

Immutables

with editing current object, you DO create the new one: int, float, str, bool, tuple, (frozenset)

Type Annotations

functionality that allows you to explicitly specify the types of variables, parameters, and return values of functions

```python
x: int = 5
name: str = "Egor"
def is_prime() -> bool:
    if len(dividers) == 2:
        return True
    return False
```

Deep

Linters

are for code static analysis: potential bugs, PEP8 violation, general recommendations

Pylint

https://github.com/pylint-dev/pylint

MyPy

https://github.com/python/mypy

Black

https://github.com/psf/black